

Some New Exact Solutions to the Dispersive Long-Wave Equation in (2+1)-Dimensional Spaces*

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Abstract In this paper, by using a further extended tanh method and symbolic computation system, some new soliton-like and period form solutions of the dispersive long-wave equation in (2+1)-dimensional spaces are obtained.

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Key words: further extended tanh method, symbolic computation system, soliton-like solutions, period form solutions, solitary waves

1 Introduction

In recent years, the studies of the exact solutions for the nonlinear evolution equations have attracted attention of many mathematicians and physicists. Many authors are interested in seeking soliton-like solutions, because the waveforms can change in different mechanisms and it usually has travelling wave solutions. In the present paper, we will study the soliton-like solutions of the dispersive long-wave equation in (2+1)-dimensional spaces^[1]

$$\begin{aligned} u_{yt} + \eta_{xx} + u_x u_y + u u_{xy} &= 0, \\ \eta_t + u_x + \eta u_x + u \eta_x + u_{xxy} &= 0. \end{aligned} \quad (1)$$

Equations (1) are firstly obtained by Boiti *et al.*^[1] as a compatibility condition for a “weak” Lax pair. In Ref. [2], Paquin and Winternitz showed that the symmetry algebra of Eq. (1) is infinite-dimensional and has a Kac–Moody–Virasoro structure. Some special similarity solutions are also given in Ref. [2] by using symmetry algebra and the classical theoretical analysis. The more general symmetry algebra, W_∞ symmetry algebra, is given in Ref. [3]. In Ref. [4], Lou gave nine types of two-dimensional similarity reductions and 13 types of ordinary differential equation reductions. In Refs. [5]–[8], Tang *et al.*, obtained the abundant localized coherent or excitation structures such as the solitons, dromions, lumps, breathers, instantons, ring solitons, peakons, compactons, localized chaotic, and fractal patterns and so on by means of the variable separation approach and defined a general type of localized excitations, folded solitary waves, and foldons. Though the model equation system is Lax or IST integrable, it does not pass the Painlevé test.^[9]

2 Exact Solutions

Recently, a direct method (extended tanh method) for finding soliton-like solutions^[10–12] has been applied suc-

cessfully to many nonlinear evolution equations and many new families of exact solutions to the dispersive long-wave equation in (2+1)-dimensional spaces have been obtained by Tian and Gao.^[11] Using this direct method, one can get not only the travelling wave solutions but also the non-travelling solitonic solutions. In Ref. [13], we introduce a further extended tanh method which can be used to construct soliton-like and period form solutions to nonlinear evolution equations.

Now we use this method to explore some new solutions of the dispersive long-wave equation in (2+1)-dimensional spaces.

To begin with, let us make a transformation of Eq. (1)

$$\eta = \pm u_y - 1, \quad (2)$$

then equation (1) is degenerated to a same differential equation,

$$u_{yt} + u_x u_y + u u_{xy} \pm u_{xxy} = 0. \quad (3)$$

We assume that equation (3) has the solution in the form

$$u = f + h\phi(px + q), \quad (4)$$

where $f = f(y, t)$, $h = h(y, t)$, $q = q(y, t)$, and $p = F_4(y)$ are arbitrary functions of y .

$$\phi' = \delta(1 + \mu\phi^2), \quad (5)$$

where δ is a nonzero constant and $\mu = \pm 1$, and the prime denotes differentiation with respect to ω , $\omega = px + q$.

Equation (5) possesses the general solution

$$\phi = \begin{cases} \tanh(\delta\omega) & \mu = -1, \\ \coth(\delta\omega) & \mu = -1, \\ \tan(\delta\omega) & \mu = 1, \\ -\cot(\delta\omega) & \mu = 1. \end{cases} \quad (6)$$

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Substituting Eqs. (4) and (5) into Eq. (3) and collecting coefficients of polynomials of ϕ and x (Notice that f ,

h , q , and p are independent of x) with the aid of Maple, then setting each coefficient to zero, we have

$$\pm 3p\mu^2\delta^2hq_y(2p\delta\mu \pm h) = 0, \quad (7)$$

$$2\mu\delta^2hp_y(q_t + fp) = 0, \quad (8)$$

$$\pm \delta p_y(2\mu\delta^2hp^2 \pm \delta h^2p \pm h_t) = 0, \quad (9)$$

$$\pm \mu\delta p_y(8\mu hp^2\delta^2 \pm 4\delta h^2p \pm h_t) = 0, \quad (10)$$

$$\delta hpf_y + f_{yt} + \delta fh_y p + \delta^2 h^2 p q_y + \delta q_t h_y + \delta h_t q_y + \delta fh p_y + \delta h q_{yt} \pm 2\delta^3 \mu h p^2 q_y = 0, \quad (11)$$

$$h_{yt} + 2\delta^2 \mu q_t h q_y + \delta h^2 p_y + 2\delta h p h_y + 2\delta^2 \mu f h p q_y \pm 2\delta^2 \mu h_y p^2 \pm 4\delta^2 \mu h p p_y = 0, \quad (12)$$

$$\pm \mu\delta(8\mu\delta^2 hp^2 q_y \pm 4\delta h^2 p q_y \pm h_y q_t \pm fh p_y \pm h q_{yt} \pm h_t q_y \pm h p f_y \pm fh_y p) = 0, \quad (13)$$

$$\pm \mu\delta(\pm 2\mu\delta fh p q_y \pm 2\mu\delta q_t h q_y + 4\mu\delta h p p_y + 2\mu\delta h_y p^2 \pm h^2 p_y \pm 2h p h_y) = 0, \quad (14)$$

It follows from Eq. (7) that $h = \pm 2\mu\delta p$, and now equations (7) $\sim$ (14) become

$$f_{yt} - 2\mu\delta^2 q_t p_y - 2\mu\delta^2 p^2 f_y - 2\mu\delta^2 p q_{yt} - 4\mu\delta^2 f p_y p = 0, \quad (15)$$

$$2\mu^2\delta^2(q_t p_y + 2f p_y p + q_{yt} p + f_y p^2) = 0, \quad (16)$$

$$-4\mu^3\delta^3 p_y p(f p + q_t) = 0. \quad (17)$$

Solving Eqs. (12) $\sim$ (14), we can obtain the following solutions easily

$$f = F_2(y) + F_1(t), \quad (18)$$

$$h = \pm 2\mu\delta F_4(y), \quad (19)$$

$$q = F_3(y) - F_4(y) \int (F_2(y) + F_1(t)) dt, \quad (20)$$

where F_1 is an arbitrary functions of t , while F_2 , F_3 , and F_4 are arbitrary functions of y .

Substituting Eqs. (18) $\sim$ (20), and (6) into Eqs. (4) and (2), we obtain the following new exact solutions of Eq. (1).

Case 1 Soliton-like solutions

Case 1a

$$\begin{aligned} u_1 &= F_2(y) + F_1(t) + 2\delta F_4(y) \tanh\left\{\delta\left[F_4(y)\left(x - \int F_1(t)dt - F_2(y)t\right) + F_3(y)\right]\right\}, \\ \eta_1 &= F_{2y}(y) + 2\delta F_{4y}(y) \tanh\left\{\delta\left[F_4(y)\left(x - \int F_1(t)dt - F_2(y)t\right) + F_3(y)\right]\right\} \\ &\quad + 2\delta^2 F_4(y) \left[F_{4y}\left(x - \int F_1(t)dt - F_2(y)t\right) + F_{3y}(y) - F_4(y)F_{2y}(y)t\right] \\ &\quad \times \operatorname{sech}^2\left\{\delta\left[F_4(y)\left(x - \int F_1(t)dt - F_2(y)t\right) + F_3(y)\right]\right\} - 1, \quad \mu = -1. \end{aligned} \quad (21)$$

Case 1b

$$\begin{aligned} u_2 &= F_2(y) + F_1(t) - 2\delta F_4(y) \tanh\left\{\delta\left[F_4(y)\left(x - \int F_1(t)dt - F_2(y)t\right) + F_3(y)\right]\right\}, \\ \eta_2 &= -F_{2y}(y) + 2\delta F_{4y}(y) \tanh\left\{\delta\left[F_4(y)\left(x - \int F_1(t)dt - F_2(y)t\right) + F_3(y)\right]\right\} \\ &\quad + 2\delta^2 F_4(y) \left[F_{4y}\left(x - \int F_1(t)dt - F_2(y)t\right) + F_{3y}(y) - F_4(y)F_{2y}(y)t\right] \\ &\quad \times \operatorname{sech}^2\left\{\delta\left[F_4(y)\left(x - \int F_1(t)dt - F_2(y)t\right) + F_3(y)\right]\right\} - 1, \quad \mu = -1. \end{aligned} \quad (22)$$

Case 1c

$$\begin{aligned} u_3 &= F_2(y) + F_1(t) + 2\delta F_4(y) \coth\left\{\delta\left[F_4(y)\left(x - \int F_1(t)dt - F_2(y)t\right) + F_3(y)\right]\right\}, \\ \eta_3 &= F_{2y}(y) + 2\delta F_{4y}(y) \coth\left\{\delta\left[F_4(y)\left(x - \int F_1(t)dt - F_2(y)t\right) + F_3(y)\right]\right\} \end{aligned}$$

$$\begin{aligned}
& + 2\delta^2 F_4(y) \left[F_{4y} \left(x - \int F_1(t) dt - F_2(y)t \right) + F_{3y}(y) - F_4(y)F_{2y}(y)t \right] \\
& \times \operatorname{csch}^2 \left\{ \delta \left[F_4(y) \left(x - \int F_1(t) dt - F_2(y)t \right) + F_3(y) \right] \right\} - 1, \quad \mu = -1.
\end{aligned} \tag{23}$$

Case 1d

$$\begin{aligned}
u_4 &= F_2(y) + F_1(t) - 2\delta F_4(y) \coth \left\{ \delta \left[F_4(y) \left(x - \int F_1(t) dt - F_2(y)t \right) + F_3(y) \right] \right\}, \\
\eta_4 &= -F_{2y}(y) + 2\delta F_{4y}(y) \coth \left\{ \delta \left[F_4(y) \left(x - \int F_1(t) dt - F_2(y)t \right) + F_3(y) \right] \right\} \\
& + 2\delta^2 F_4(y) \left[F_{4y} \left(x - \int F_1(t) dt - F_2(y)t \right) + F_{3y}(y) - F_4(y)F_{2y}(y)t \right] \\
& \times \operatorname{csch}^2 \left\{ \delta \left[F_4(y) \left(x - \int F_1(t) dt - F_2(y)t \right) + F_3(y) \right] \right\} - 1, \quad \mu = -1.
\end{aligned} \tag{24}$$

Remark We call it a solitary wave solution if a certain solution contains the variable $p(y, t)x + q(y, t)$, i.e., a linear form of x, y and t ; otherwise, we call it a soliton-like solution.

It is easy to see that the solutions (34) and (35) in Ref. [11] are just some special cases of Case 1a, i.e., $F_1 = 0$, $F_2 = \theta$, $F_3 = ay + c$, $\delta = 1/2$ here. But the other solutions that we obtained in this paper were not obtained by using that method.^[10–12]

Case 2 Period form solutions**Case 2a**

$$\begin{aligned}
u_5 &= F_2(y) + F_1(t) - 2\delta F_4(y) \tan \left\{ \delta \left[F_4(y) \left(x - \int F_1(t) dt - F_2(y)t \right) + F_3(y) \right] \right\}, \\
\eta_5 &= F_{2y}(y) - 2\delta F_{4y}(y) \tan \left\{ \delta \left[F_4(y) \left(x - \int F_1(t) dt - F_2(y)t \right) + F_3(y) \right] \right\} \\
& - 2\delta^2 F_4(y) \left[F_{4y} \left(x - \int F_1(t) dt - F_2(y)t \right) + F_{3y}(y) - F_4(y)F_{2y}(y)t \right] \\
& \times \sec^2 \left\{ \delta \left[F_4(y) \left(x - \int F_1(t) dt - F_2(y)t \right) + F_3(y) \right] \right\} - 1, \quad \mu = 1.
\end{aligned} \tag{25}$$

Case 2b

$$\begin{aligned}
u_6 &= F_2(y) + F_1(t) + 2\delta F_4(y) \tan \left\{ \delta \left[F_4(y) \left(x - \int F_1(t) dt - F_2(y)t \right) + F_3(y) \right] \right\}, \\
\eta_6 &= -F_{2y}(y) - 2\delta F_{4y}(y) \tan \left\{ \delta \left[F_4(y) \left(x - \int F_1(t) dt - F_2(y)t \right) + F_3(y) \right] \right\} \\
& - 2\delta^2 F_4(y) \left[F_{4y} \left(x - \int F_1(t) dt - F_2(y)t \right) + F_{3y}(y) - F_4(y)F_{2y}(y)t \right] \\
& \times \sec^2 \left\{ \delta \left[F_4(y) \left(x - \int F_1(t) dt - F_2(y)t \right) + F_3(y) \right] \right\} - 1, \quad \mu = 1.
\end{aligned} \tag{26}$$

Case 2c

$$\begin{aligned}
u_7 &= F_2(y) + F_1(t) + 2\delta F_4(y) \cot \left\{ \delta \left[F_4(y) \left(x - \int F_1(t) dt - F_2(y)t \right) + F_3(y) \right] \right\}, \\
\eta_7 &= F_{2y}(y) + 2\delta F_{4y}(y) \cot \left\{ \delta \left[F_4(y) \left(x - \int F_1(t) dt - F_2(y)t \right) + F_3(y) \right] \right\} \\
& - 2\delta^2 F_4(y) \left[F_{4y} \left(x - \int F_1(t) dt - F_2(y)t \right) + F_{3y}(y) - F_4(y)F_{2y}(y)t \right] \\
& \times \csc^2 \left\{ \delta \left[F_4(y) \left(x - \int F_1(t) dt - F_2(y)t \right) + F_3(y) \right] \right\} - 1, \quad \mu = 1.
\end{aligned} \tag{27}$$

Case 2d

$$\begin{aligned}
u_8 &= F_2(y) + F_1(t) - 2\delta F_4(y) \cot \left\{ \delta \left[F_4(y) \left(x - \int F_1(t) dt - F_2(y)t \right) + F_3(y) \right] \right\}, \\
\eta_7 &= -F_{2y}(y) + 2\delta F_{4y}(y) \cot \left\{ \delta \left[F_4(y) \left(x - \int F_1(t) dt - F_2(y)t \right) + F_3(y) \right] \right\}
\end{aligned}$$

$$\begin{aligned}
& -2\delta^2 F_4(y) \left[F_{4y} \left(x - \int F_1(t) dt - F_2(y)t \right) + F_{3y}(y) - F_4(y)F_{2y}(y)t \right] \\
& \times \csc^2 \left\{ \delta \left[F_4(y) \left(x - \int F_1(t) dt - F_2(y)t \right) + F_3(y) \right] \right\} - 1, \quad \mu = 1.
\end{aligned} \tag{28}$$

It is obvious that we have obtained some new soliton-like and period form solutions that are not obtained by the previously known method.

3 Conclusions

In summary, we have obtained some new soliton-like and period form solutions for the dispersive long-wave equation in (2+1)-dimensional spaces by using a simple transformation (2) and the further extended tanh method.

The method can also be applied to other nonlinear coupled evolution equations such as the integrable Broer–Kaup equations in (2+1)-dimensional spaces. The result will be given in another paper.

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